

CANDIDATE
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CANDIDATE
NUMBER

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0893/01

April 2023

45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should show all your working in the booklet.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **20** pages. Any blank pages are indicated.



1 This question is about the human excretory (renal) system.

(a) Complete these sentences about the structure and function of the excretory (renal) system.

The excretory (renal) system filters blood.

Blood enters the where it is filtered, and urine is formed.

This urine passes along a tube called the towards the bladder.

The urine is stored in the bladder.

Urine is released from the body through a different tube called the

[3]

(b) The table shows the percentage concentration of four substances in blood plasma and in urine.

substance	percentage concentration of substance	
	in blood plasma	in urine
ammonia	0.00	0.05
protein	9.00	0.00
salt	0.60	0.90
urea	0.03	2.00

Which substance has the **greatest increase** in the urine compared to blood plasma?

Choose from the list.

ammonia

protein

salt

urea

..... [1]

2 Aiko calculates the density of some objects.

(a) A block of iron has a mass of 19.68 g.

The volume of the block is 2.50 cm^3 .

(i) Write down the equation Aiko uses to calculate density.

density = [1]

(ii) Calculate the density of Aiko's block of iron.

density = g/cm^3 [1]

(b) Aiko calculates the density of four objects.

Look at her results.

object	density in g/cm^3
A	2.7
B	8.4
C	0.002
D	13.6

Which object is a gas?

Circle the correct answer.

A

B

C

D

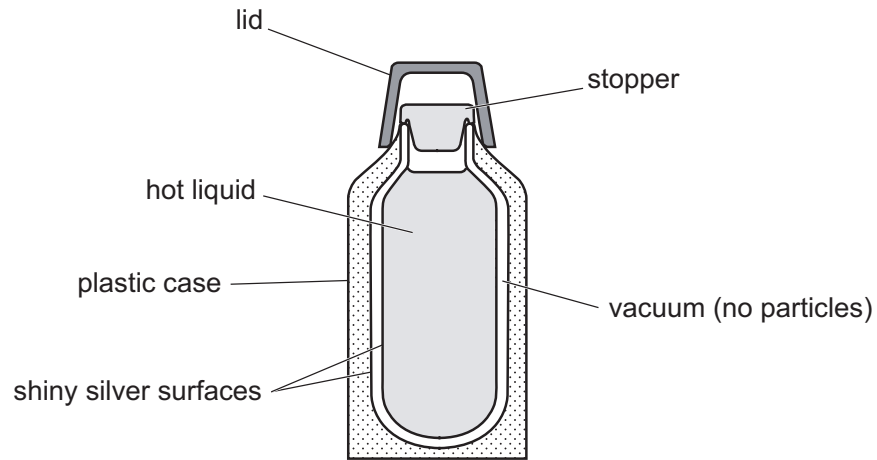
Explain your answer.

..... [1]

- 3 Look at the diagram of a vacuum flask.

Gabriella puts a hot liquid into the vacuum flask.

The flask keeps the hot liquid warm.



- (a) Which material is most suitable to make the stopper?

Circle the correct answer.

copper

gold

iron

plastic

[1]

- (b) (i) Suggest how the shiny silver surfaces help to keep the liquid warm.

..... [1]

- (ii) The vacuum does **not** contain any particles.

Explain why the vacuum reduces the transfer of thermal energy.

.....

 [2]

(c) Gabriella removes the lid and the stopper from the vacuum flask.

Some of the hot liquid evaporates.

What happens to the temperature of the liquid that remains in the vacuum flask?

Explain why.

Use ideas about particles.

.....

.....

..... [3]

4 The Earth's crust is made of a number of large pieces.

(a) What is the name given to these pieces of the Earth's crust?

Circle the correct answer.

earthquake

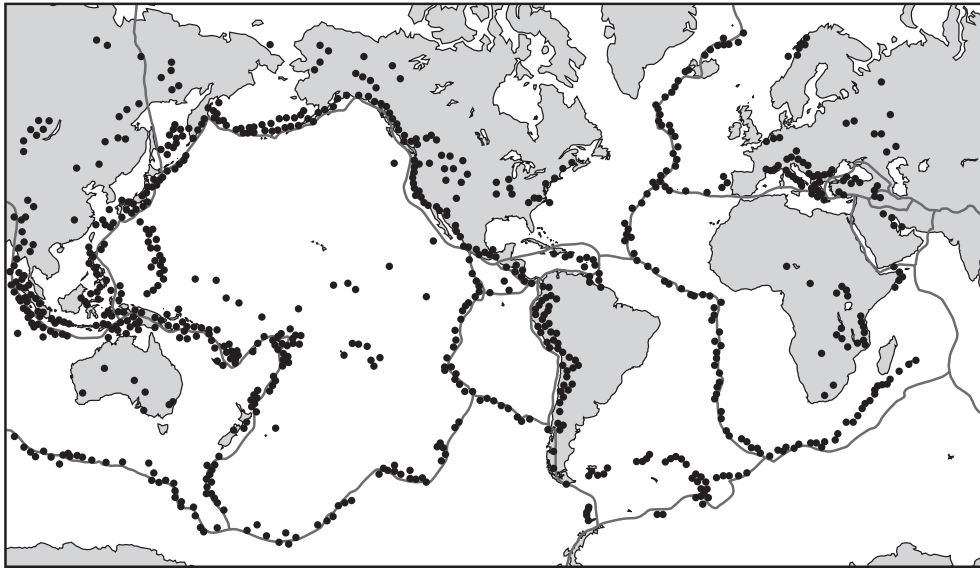
inner core

mantle

tectonic plates

[1]

(b) Look at the map showing the positions of earthquakes around the world in 2021.

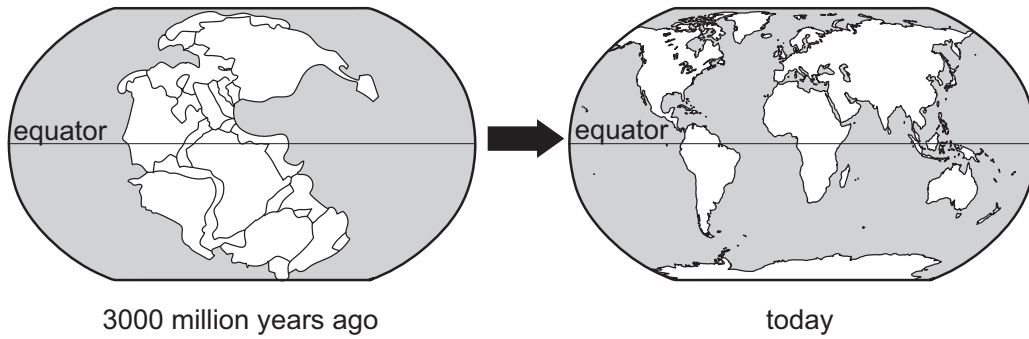


• earthquake

Suggest how the position of earthquakes provides evidence for the large pieces of the crust.

.....
..... [1]

(c) Look at the two maps showing the jigsaw appearance of the continental coasts.



(i) What is meant by the **jigsaw appearance** of the continental coasts?

.....
 [1]

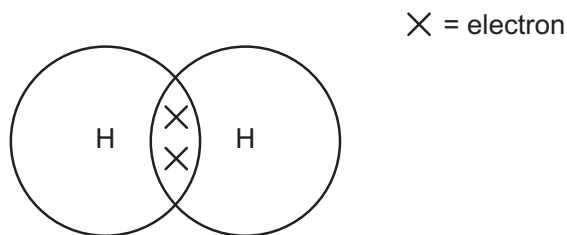
(ii) The continents have moved since 3000 million years ago.

Explain how the continents move.

.....

 [2]

- 5 The diagram shows a model of a hydrogen molecule.



- (a) Name the **type** of bonding in a hydrogen molecule.

.....

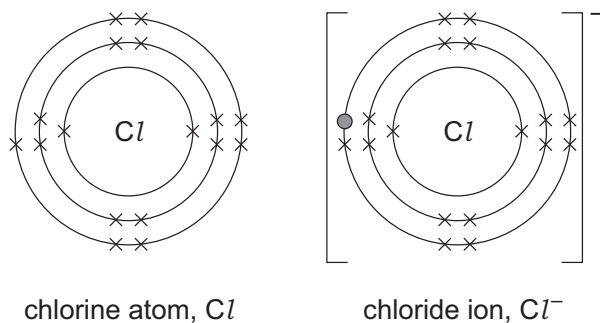
Explain how you can tell from the diagram.

.....

.....

[2]

- (b) Look at the diagrams showing a chlorine atom and a chloride ion.



Describe how a chloride ion is made from a chlorine atom.

..... [1]

- (c) Sodium chloride, $NaCl$, is made up of sodium ions, Na^+ , and chloride ions, Cl^- .

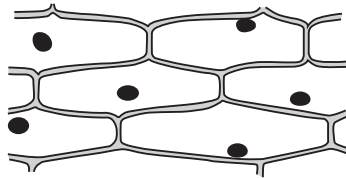
Explain how the ions in sodium chloride are held together.

.....

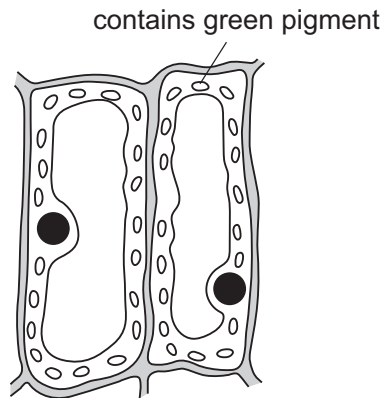
..... [1]

6 This is a question about photosynthesis and plant minerals.

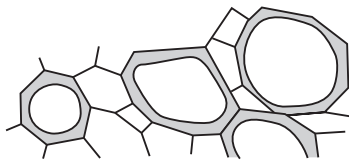
(a) Look at the diagrams of different plant cells.



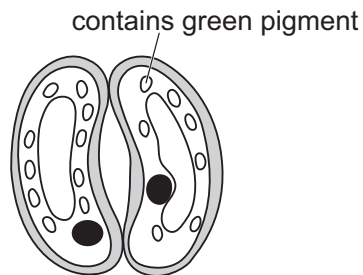
A onion epidermal cells



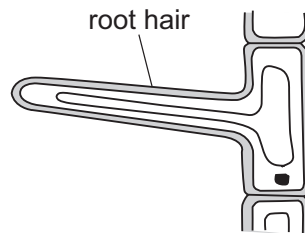
B palisade cells



C cells found in xylem



D guard cells



E epidermal cell in the root

Which **two** diagrams show plant cells that photosynthesise?

Choose from **A**, **B**, **C**, **D** and **E**.

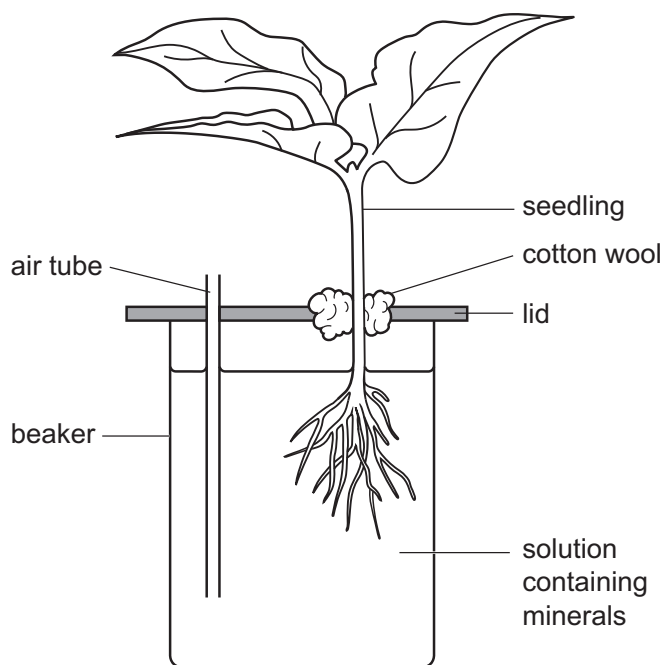
..... and

[2]

(b) Carlos investigates the effect of magnesium on plant growth.

Carlos makes a hypothesis about the effect of magnesium on plant growth.

He uses the equipment in the diagram.



In his first experiment Carlos:

- fills a beaker with a solution containing all the minerals needed for healthy growth
- assembles the equipment and seedling as shown in the diagram
- records the appearance of the seedling after four weeks.

Carlos repeats the experiment.

In his second experiment, he uses a solution that contains all the minerals needed for healthy growth **except** magnesium.

(i) Carlos makes a hypothesis about the effect of magnesium on plant growth.

Suggest a suitable hypothesis.

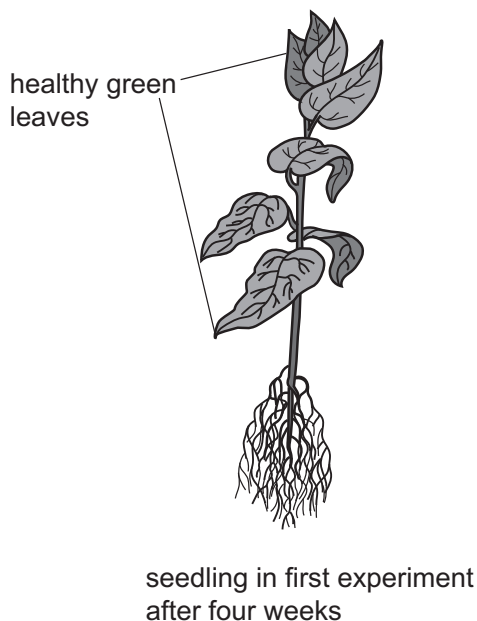
.....
..... [1]

- (ii) In his first experiment, Carlos uses a solution with all the minerals needed for healthy growth.

Explain why this is important.

.....
 [1]

- (c) Carlos draws a diagram to show the appearance of the plant from the first experiment after four weeks.



Carlos repeats his experiment again.

In his third experiment, he uses a solution that contains all the minerals needed for healthy growth **except** nitrates.

Predict the appearance of the seedling left in the solution **without** nitrates after four weeks.

Give a reason for your answer.

.....

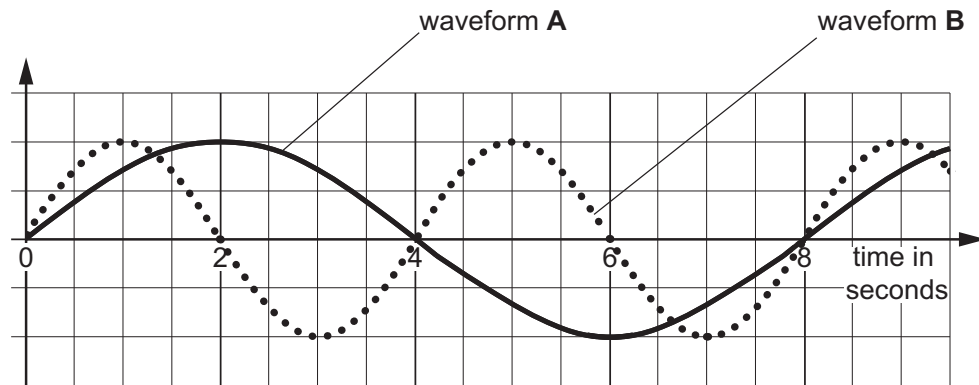
 [2]

- (d) Which substance is made inside chloroplasts?

..... [1]

7 Ahmed hears two sounds, **A** and **B**.

Look at the waveforms for these two sounds.



Give **one similarity** and **one difference** between waveform **A** and waveform **B**.

similarity

difference

[2]

- 8 When asteroids collide with the Earth, they make craters.

Chen uses a model to investigate the effect of asteroid collisions with the Earth.

In his investigation Chen:

- drops a rock into a container of soil
- measures the size of the hole in the soil made by the rock.

- (a) Complete the sentences about Chen's model.

In the model the rock represents

In the model the soil represents

In the model the hole represents

[2]

- (b) Write down **one strength** and **one limitation** of his model of asteroid collisions with the Earth.

strength

.....

limitation

.....

[2]

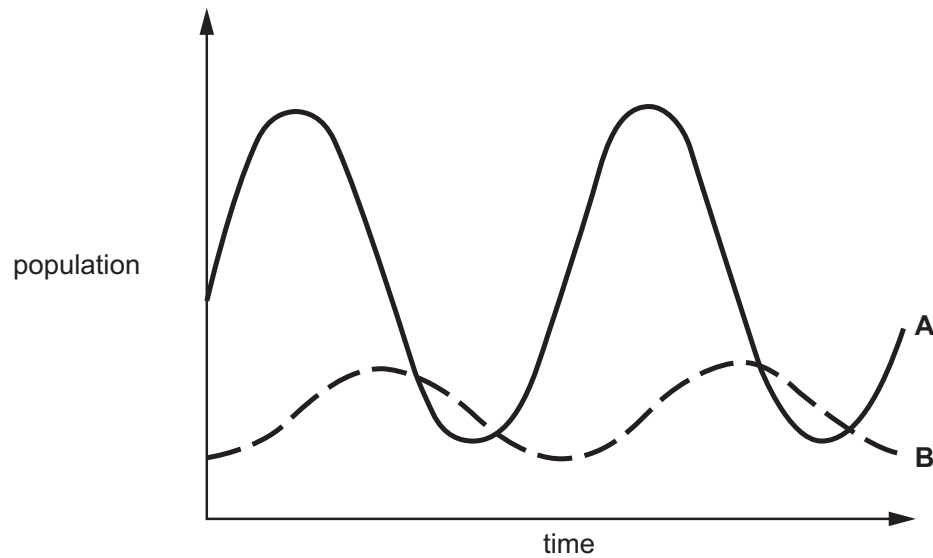
9 Myxomatosis is a disease that kills rabbits.

(a) Describe the effect of myxomatosis on the size of a rabbit population.

..... [1]

(b) Foxes hunt rabbits for food.

The graph shows the population of foxes and the population of rabbits.



Tick (✓) the box that shows the population of rabbits.

line A

☐

line B

☐

Give **two** reasons for your answer.

1

.....

2

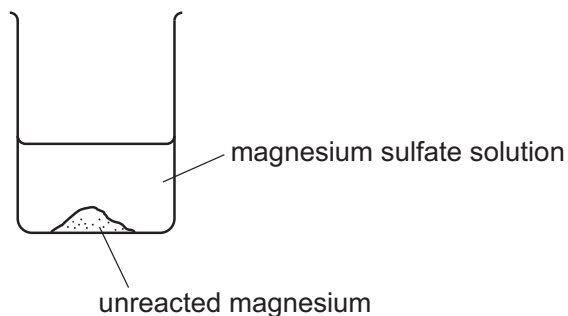
.....

[2]

10 Blessy makes some magnesium sulfate.

She adds an excess of magnesium to some dilute sulfuric acid until the reaction stops.

Unreacted magnesium is left at the bottom of the solution.



(a) Describe how Blessy separates the magnesium sulfate solution from the unreacted magnesium.

..... [1]

(b) Blessy wants to make solid magnesium sulfate from the magnesium sulfate solution.

Describe how she makes solid magnesium sulfate.

.....
..... [1]

(c) Blessy now wants to make zinc chloride.

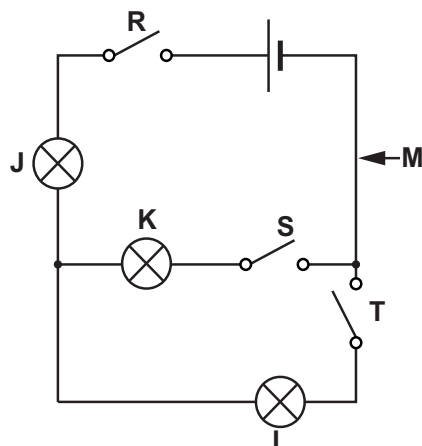
Write down the name of the **metal** and **acid** she uses to make zinc chloride.

metal

acid

[1]

11 The circuit diagram shows the circuit Priya makes using switches and identical lamps.



(a) Priya connects a meter to measure the current at position **M** in the circuit.

Draw the correct symbol for the meter she uses to measure the current.

..... [1]

(b) Priya opens and closes different switches.

Complete the table by writing if the:

- switches are **open** or **closed**
- lamps are **on** or **off**.

switch R	switch S	switch T	lamp J	lamp K	lamp L
closed	open	open	off		
open	closed	closed		off	
.....			on	off	on

[3]

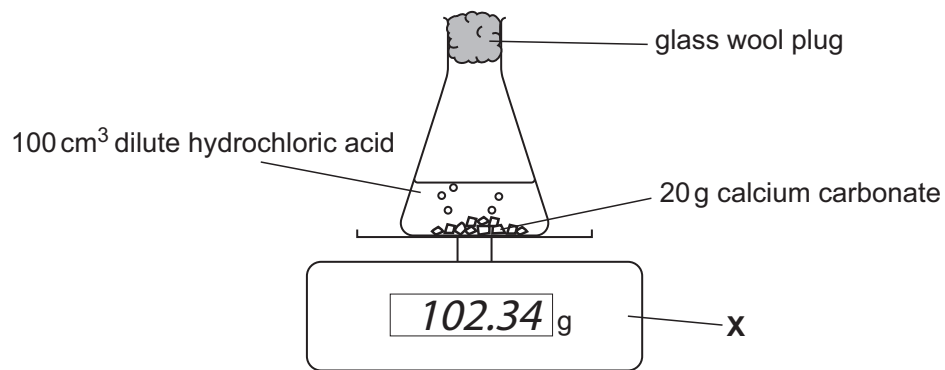
(c) Describe how Priya connects a voltmeter to measure the voltage across lamp **J**.

..... [1]

12 Pierre investigates the reaction between calcium carbonate and dilute hydrochloric acid.

The reaction gives off carbon dioxide gas.

Look at the equipment he uses.

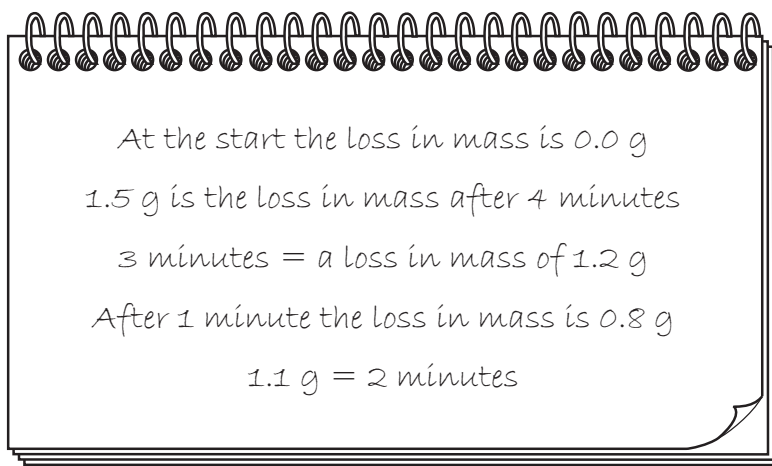


(a) Write down the name of equipment **X**.

..... [1]

(b) Pierre measures the loss in mass every minute for 4 minutes.

Here are his results.



Complete his results table.

	loss in mass in g
.....	
.....	
.....	
.....	
.....	
.....	
.....	

[2]

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The Periodic Table of Elements

Group																													
1	2											3	4	5	6	7	8												
		Key atomic number atomic symbol name relative atomic mass												1 H hydrogen 1															
3 Li lithium 7	4 Be beryllium 9													5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20										
11 Na sodium 23	12 Mg magnesium 24													13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40										
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84												
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xeron 131												
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —												
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganeson —												

lanthanoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium –	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium –	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium –	94 Pu plutonium –	95 Am americium –	96 Cm curium –	97 Bk berkelium –	98 Cf californium –	99 Es einsteinium –	100 Fm fermium –	101 Md mendelevium –	102 No nobelium –	103 Lr lawrencium –

actinoids



Cambridge Lower Secondary Checkpoint

SCIENCE

0893/01

Paper 1

April 2023

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Markers were instructed to award marks. It does not indicate the details of the discussions that took place at a Markers' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the End of Series Report. Cambridge will not enter into discussions about these mark schemes.

This document has **16** pages.

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed so answers do not need to be grammatically correct. Significant figures will be indicated in the question or in the mark scheme.

Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>Underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working, provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

Question	Answer	Marks	Further Information
1(a)	kidney ureter urethra	3	each correct answer = 1 mark Accept nephron Accept ureta or uretur Do not accept uretra or uretha or urether or urater or urethera Accept ureathra or urethera Do not accept urether or uretha or urathra
1(b)	urea	1	Accept any indication of correct answer, e.g. ticking, underlining or circling, but answer line takes precedence

Question	Answer	Marks	Further Information
2(a)(i)	(density =) $\frac{\text{mass}}{\text{volume}}$	1	Accept mass ÷ volume Accept m/v Ignore g / cm ³
2(a)(ii)	7.87 (g / cm ³)	1	Accept 7.9 or 7.872 Ignore any unit in answer Note no ecf from (a)(i)
2(b)	C and has a (very) low density	1	both correct for the mark Accept any indication of correct answer, e.g. ticking, underlining or in explanation, but circling takes precedence Accept has the lowest density Note a density below 1 is not sufficient Note it is the lowest is not sufficient – must mention density or dense in answer Ignore it is the lightest

Question	Answer	Marks	Further Information
3(a)	plastic	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence
3(b)(i)	(idea that) shiny surface reflects thermal energy back into the liquid	1	Accept they are good reflectors (of thermal energy) Accept reflect radiation Ignore reflects light Ignore reference to absorbing Ignore silver is a good conductor
3(b)(ii)	convection cannot take place conduction cannot take place	2	each correct answer = 1 mark Accept conduction and convection in any order Do not accept radiation cannot take place max 1 mark for the question

Question	Answer	Marks	Further Information
3(c)	(liquid) cools down or (temperature) decreases high(er) energy particles (near the surface of the liquid) escape (into the air) (average) energy of the remaining particles decreases	3	each correct answer = 1 mark Accept (idea that) temperature becomes low or becomes colder or results in a low temperature Note temperature is cold is not sufficient Accept fast(er) moving particles escape or high(er) energy particles evaporate Accept particles that evaporate take some energy with them or particles use energy to evaporate Do not accept answers that refer to heat particles having more energy or moving faster Accept energy of the particles in the liquid decreases or slow moving particles left in liquid or low energy particles left in liquid Accept evaporation is an endothermic process as additional marking point Ignore ideas about conduction and convection Ignore answers that refer to hot particles and cold particles

Question	Answer	Marks	Further Information
4(a)	tectonic plates	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence
4(b)	(the earthquakes) occur at the plate boundaries or (the earthquakes) happen at the edges of the pieces of crust	1	Accept earthquakes happen when the plates collide or occur at places where tectonic plates meet Ignore earthquakes happen at fault lines
4(c)(i)	(idea that) the coastlines fit together (like a jigsaw puzzle)	1	Accept coastlines have a complementary shape or coastlines combine together like a jigsaw puzzle Accept continents fit together or tectonic plates fit together Ignore after many years the continents no longer fit together Ignore countries fit together Do not accept have the same shape
4(c)(ii)	any two from magma moves or magma flows convection currents (in the mantle) (convection currents or magma movements) drag/pull the continents or drag/pull the tectonic plate (apart or together)	2	each correct answer = 1 mark Ignore continents move Accept convection currents in mantle or convection currents of magma = 2 marks Note tectonic plates move is not sufficient

Question	Answer	Marks	Further Information
5(a)	covalent (pair of) shared electrons	2	each correct answer = 1 mark Note ionic bond = 0 marks for the question
5(b)	(chlorine atom) gains one electron (to make a chloride ion)	1	Note assume answer refers to a chlorine atom Accept gains an electron Do not accept chloride gains an electron
5(c)	attraction between positive and negative ions	1	Accept electrostatic attraction (between ions) Note by an ionic bond is not sufficient Do not accept reference to intermolecular forces

Question	Answer	Marks	Further Information
6(a)	B (and) D	2	each correct answer = 1 mark Accept either order Accept palisade (and) guard cell, but letter takes precedence Accept any indication of the correct answer, e.g. ticking, underlining or circling on the diagrams, but answer line takes precedence
6(b)(i)	any one from magnesium will make plants green or magnesium makes the plant's green pigment or magnesium will make chlorophyll lack of magnesium will reduce plant growth lack of magnesium will result in yellow (leaves)	1	Note answers must specifically refer to either the presence or the absence of magnesium Accept ora Accept if magnesium not present the plants are not green Ignore without magnesium there is no photosynthesis or without magnesium the rate of photosynthesis decreases
6(b)(ii)	(acts as a) control or (idea of) to be able to compare (the seedlings with or without specific nutrients)	1	Accept (idea of) to see the differences

Question	Answer	Marks	Further Information
6(c)	(idea of) smaller in size or reduced growth or fewer leaves (idea of) less protein made	2	each correct answer = 1 mark Accept less nitrogen to make protein or no nitrates to make protein or less amino acids made Ignore plants will die or no growth Ignore reference to colour of the leaves Ignore reference to carbohydrate
6(d)	carbohydrate or glucose	1	Accept oxygen or starch Accept chlorophyll

Question	Answer	Marks	Further Information
7	(similarity) they have the same amplitude (difference) waveform B has a higher frequency than waveform A / ora	2	each correct answer = 1 mark Accept they have the same loudness or same volume Accept waveform B has a higher pitch (than waveform A)/ ora Accept waveform B has a shorter wavelength (than waveform A)/ ora

Question	Answer	Marks	Further Information
8(a)	(rock) asteroid (soil) Earth or crust (hole) crater	2	all three correct = 2 marks one or two correct = 1 mark
8(b)	(strength) any one from helps to visualize asteroid collisions which you cannot see normally can investigate relationship between size/speed/height of asteroid and crater size can investigate relationship between size/speed/height of asteroid and distance travelled by debris (idea that) it is very easy to do (idea that) it is easy to repeat (idea that) it shows how craters are formed	2	correct strength = 1 mark Accept (idea that) provides an understanding about what happens during an asteroid collision Accept the model is easy to understand Accept the model can be used with different sizes of rocks/asteroids Accept can see what happens when an asteroid hits the Earth surface Accept simulates a real event Accept can accurately represent an asteroid collision

Question	Answer	Marks	Further Information
8(c)	(limitation) any one from difficult to make it a fair test, e.g. rocks not all the same size/mass rocks may hit the bottom of the container or soil may not be deep enough rocks are (much) smaller than real asteroids or forces are (much) smaller than for real asteroids cannot represent the real speed of an asteroid hitting the Earth does not show other consequences of impact, e.g. dust formation the Earth's surface is not just made of soil	2	correct limitation = 1 mark Accept (idea that) model cannot be done on a large scale Accept (idea that) does not investigate what happens when an asteroid hits oceans Accept (idea that) cannot tell you about a collision of a large asteroid Accept does not (fully) represent the force at which an asteroid hits the Earth Accept (idea that) there is a limited drop height Ignore it is inaccurate is not sufficient but accept inaccurate that is qualified, e.g. cannot be scaled up accurately

Question	Answer	Marks	Further Information
9(a)	(causes) a decrease or decline	1	Accept it goes down or drops or falls Ignore extinction or killed
9(b)	A (no mark) any two from smaller population size of foxes compared to rabbits larger population rise seen at the start in rabbit or prey when population of foxes decreases, the rabbit population will increase or when population of rabbit declines, the population of foxes will also decline	2	B ticked = 0 marks for the question each correct answer = 1 mark Accept change in line A is bigger than change in line B Accept ora Accept ora Accept (idea that) fox population rises after rabbit population rises = 2 marks Accept (idea that) predator population rises after prey population rises = 2 marks Accept (idea that) fox population declines after rabbit population declines = 2 marks Accept (idea that) predator population declines after prey population declines = 2 marks Accept change in line B is always after change in line A = 2 marks

Question	Answer	Marks	Further Information
10(a)	filtration or filtering	1	Accept use filter paper Do not accept sieving or strainer
10(b)	evaporation or evaporate the mixture	1	Accept heat or heating or boil off water Accept leave in a hot place Ignore crystallization
10(c)	(metal) zinc (acid) (dilute) hydrochloric acid	1	both correct for the mark Accept Zn or HCl but name takes precedence

Question	Answer	Marks	Further Information																								
11(a)		1	Accept symbol drawn on diagram, but answer space takes precedence if symbol drawn there Do not accept a line through the ammeter symbol																								
11(b)	<table><tr><th>switch R</th><th>switch S</th><th>switch T</th><th>lamp J</th><th>lamp K</th><th>lamp L</th></tr><tr><td>(closed)</td><td>(open)</td><td>(open)</td><td>(off)</td><td>off</td><td>off</td></tr><tr><td>(open)</td><td>(closed)</td><td>(closed)</td><td>off</td><td>(off)</td><td>off</td></tr><tr><td>closed</td><td>open</td><td>closed</td><td>(on)</td><td>(off)</td><td>(on)</td></tr></table>	switch R	switch S	switch T	lamp J	lamp K	lamp L	(closed)	(open)	(open)	(off)	off	off	(open)	(closed)	(closed)	off	(off)	off	closed	open	closed	(on)	(off)	(on)	3	each correct row = 1 mark
switch R	switch S	switch T	lamp J	lamp K	lamp L																						
(closed)	(open)	(open)	(off)	off	off																						
(open)	(closed)	(closed)	off	(off)	off																						
closed	open	closed	(on)	(off)	(on)																						
11(c)	in parallel (with lamp J)	1	Accept annotations added to the diagram may help the award of the mark Accept description of parallel e.g. need to connect one end of the meter before the lamp and one after the lamp																								

Cambridge Lower Secondary Checkpoint

CANDIDATE
NAME

CENTRE
NUMBER

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CANDIDATE
NUMBER

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SCIENCE

0893/02

Paper 2

April 2023

45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should show all your working in the booklet.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **20** pages. Any blank pages are indicated.

1 This question is about a small mammal called a shrew.

(a) Look at the picture of two shrews from the same species.



Each shrew has a different fur colour.

Fur colour is controlled by genes.

Write down the name of the chemical that makes up genes.

[1]

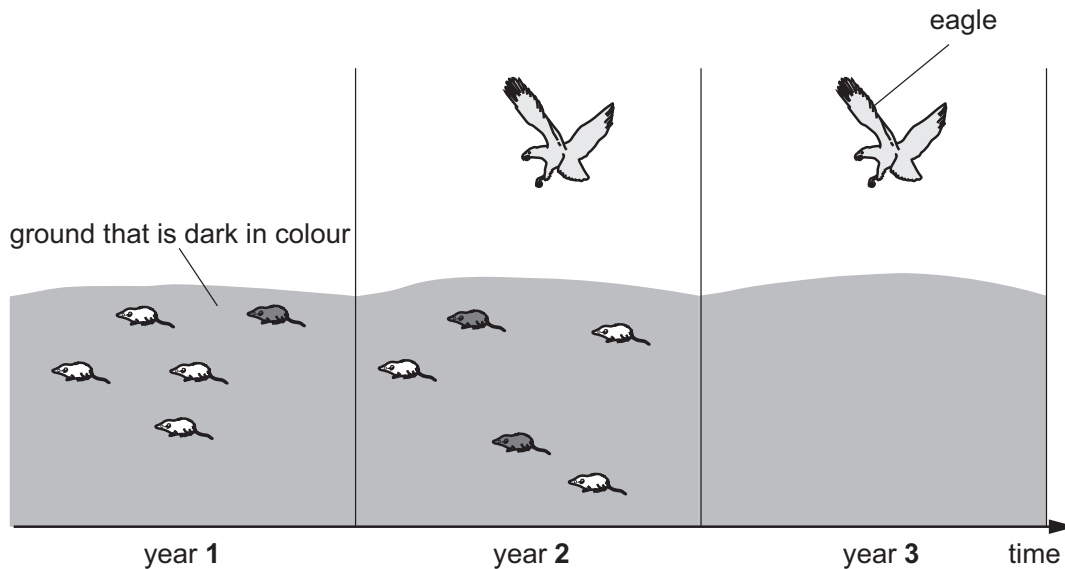
(b) Eagles feed on shrews.

Scientists sample the population of the shrews living on ground that is dark in colour.

The diagram shows the population of shrews over two years.

The population of shrews for year 3 is **not** shown.

An eagle feeds on the shrews during years 2 and 3.



- (i) The total population size of the shrews is the same during all three years.

Predict the numbers of white shrews and grey shrews in year 3.

Write your prediction in the table.

fur colour	number in population in year 3
	
	

[1]

- (ii) Describe how natural selection explains the changes in the numbers of white and grey shrews.

.....

.....

.....

..... [3]

- (iii) The scientists had to trap and release the shrews to get their results.

Write down **two** safety precautions the scientists took when trapping and releasing the shrews.

1

2

[2]

3 Water moves through a plant.

(a) The table shows information about the pathway of water into and out of a plant.

Number **1** is the part where water enters the plant.

Number **5** is the part where water is lost from the plant.

Complete the table by writing the numbers **1, 2, 3, 4** and **5** to show the pathway of water into and out of a plant.

One has been done for you.

part of plant	order of pathway
leaf	5
leaf xylem	
root hair cell	
root xylem	
stem xylem	

[1]

(b) Plants lose water from the surface of their leaves.

Write down the name of this process.

..... [1]

(c) A desert plant grows well due to its very waxy leaves.

High carbon dioxide levels reduce wax production in these plants.

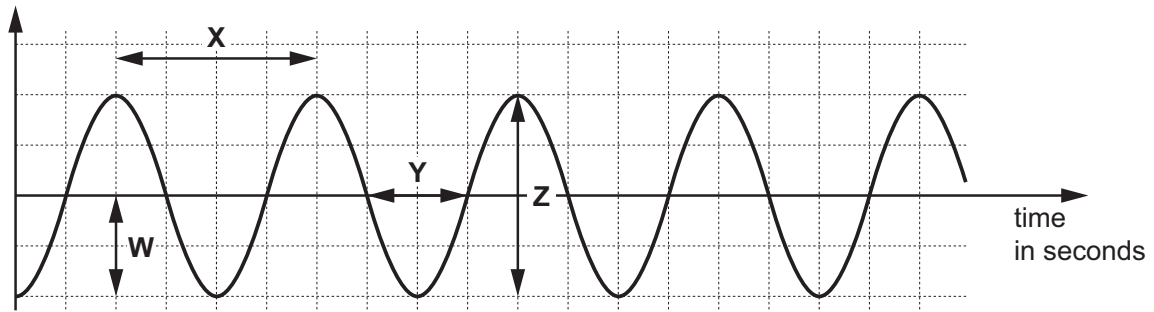
Cars produce carbon dioxide.

Suggest what happens to the population of these desert plants growing near a new road.

.....
 [1]

4 This question is about waveforms.

(a) Look at the diagram of the waveform of a sound.

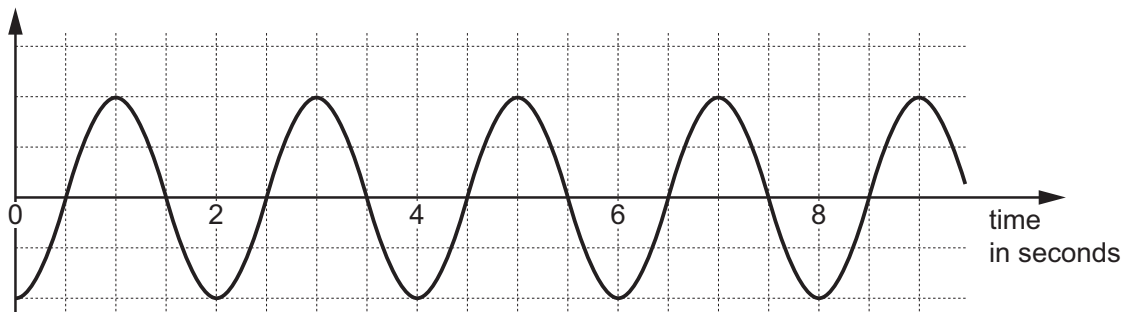


Which letter shows the amplitude of the sound wave?

Circle the correct answer.

W X Y Z [1]

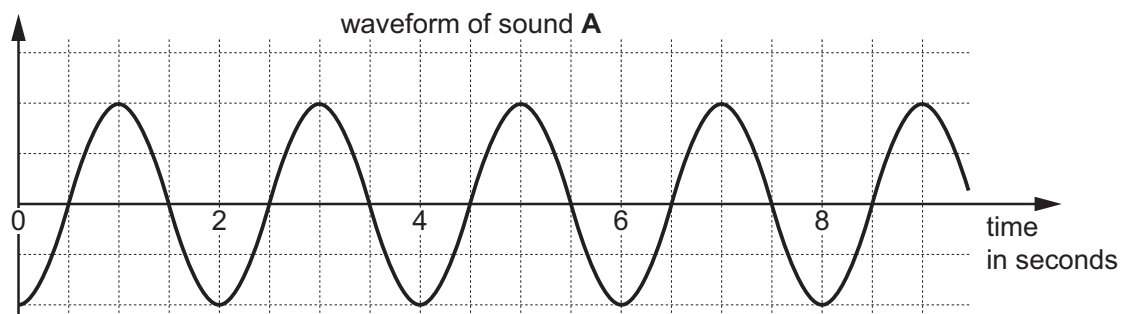
(b) The waveform for the sound shows several waves.



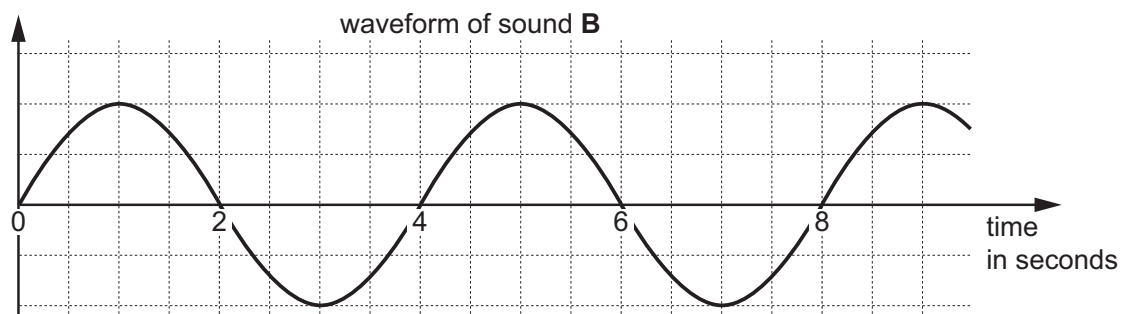
How many **complete** waves are there in 8 seconds of the waveform?

number of waves [1]

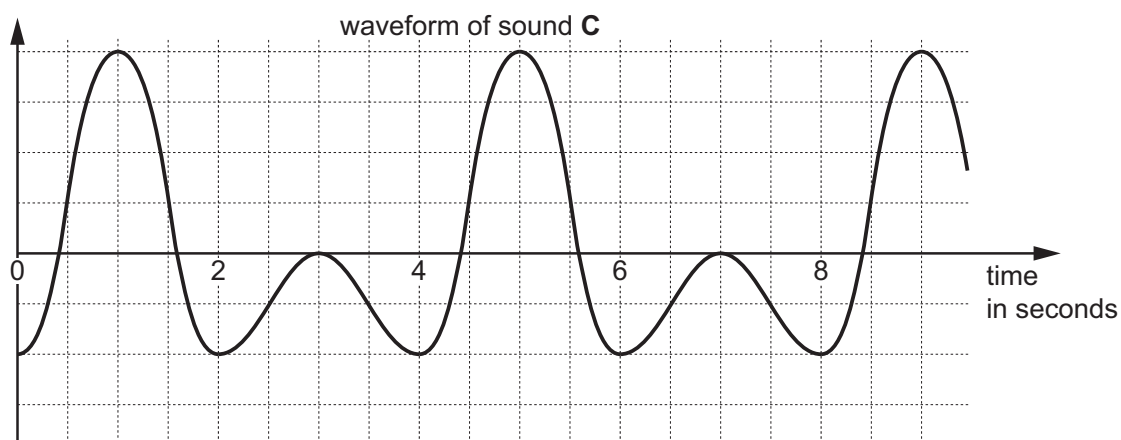
(c) The diagrams show how sound **A** interacts with sound **B** to make sound **C**.



+



=



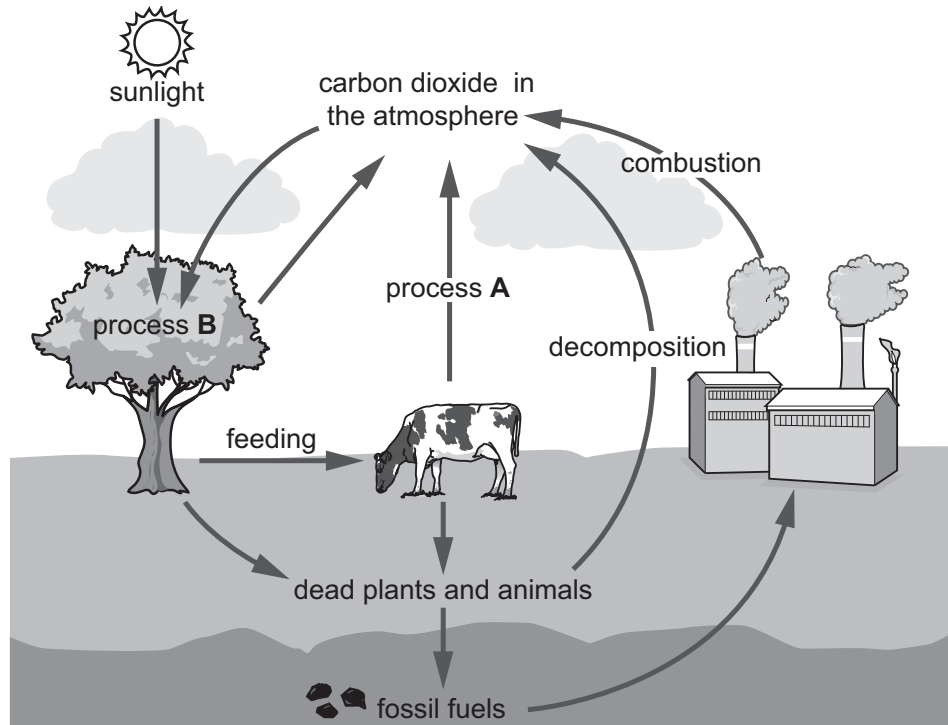
Describe how the waveform of sound **A** and waveform of sound **B** interact at:

time = 1 second

time = 3 seconds

[2]

- 5 Look at the diagram of the carbon cycle.



- (a) Write down the name of process A.

..... [1]

- (b) Write down the name of process B.

..... [1]

- (c) Write down **one similarity** between combustion and decomposition.

..... [1]

- (d) Scientists believe that too much carbon dioxide in the atmosphere causes climate change.

Describe **two** possible effects of climate change.

1

2

.....

[2]

6 Look at the data about some Group 1 elements.

element	melting point in °C	atomic radius in pm
lithium	181	145
sodium	98	180
potassium	64	220
rubidium		235

(a) Describe the trend in atomic radius as you go down Group 1.

.....
 [1]

(b) Predict the melting point of rubidium.

The melting point of rubidium is °C. [1]

(c) Sodium reacts with chlorine to make an ionic solid called sodium chloride.

Sodium chloride has a melting point of 808 °C.

Sodium chloride has a structure.

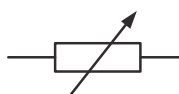
Name this type of structure.

..... [1]

7 Write down the names of these electrical symbols.



.....



.....

[2]

- 8 Here are some sentences about the collision theory for the formation of the Moon.

A	The less dense rocks eventually merged together to form the Moon.
B	This caused very high temperatures and the Earth's outer layer melted.
C	A collision occurred between the Earth and a small planet.
D	The less dense rocks were ejected and cooled.
E	The dense iron from the cores of both planets merged to create the Earth.
F	The less dense rocks were captured by the Earth's gravitational field.

- (a) Put these sentences in the correct order to describe the collision theory.

Two have been done for you.

		E		F	
--	--	---	--	---	--

[3]

- (b) In 1969, astronauts went to the Moon and collected rock samples.

Describe how these Moon rock samples provide evidence to support the collision theory.

.....

..... [1]

- (c) Suggest why the density of the Earth is greater than the density of the Moon.

Use ideas from the collision theory in your answer.

.....

..... [1]

- 9 Scientists estimate one million species of plants and animals are at risk of extinction.

(a) Which factors cause a species to become extinct?

Tick (✓) the **two** correct factors.

changes in seasons

☐

changes to the environment over time

☐

increased reproduction

☐

new diseases

☐

new food sources

☐

[2]

(b) Explain what is meant by the statement:

‘An animal is at risk of extinction.’

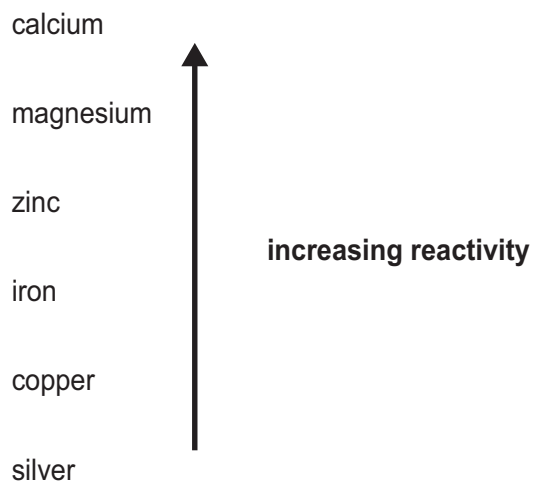
Use ideas about death rate and reproduction rate.

.....

..... [1]

10 Look at the list of metals in order of reactivity.

The most reactive metal is at the top.



A piece of copper is dipped into silver nitrate solution.

Silver is made.

A blue solution of copper nitrate is also made.

(a) What is the name of this **type** of reaction?

Circle the correct answer.

crystallisation

decomposition

displacement

filtration

neutralisation

[1]

(b) Write the word equation for this reaction.

..... [1]

(c) Look at the list of metals and solutions.

Tick (✓) to show if the metal reacts with the solution.

copper + iron nitrate

☐

magnesium + zinc nitrate

☐

silver + magnesium nitrate

☐

iron + zinc nitrate

☐

Explain your answer.

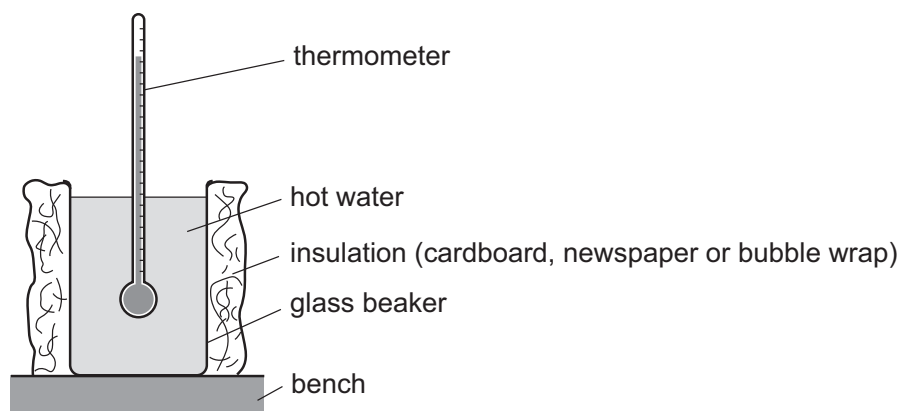
.....

.....

..... [2]

- 11 Lily investigates which type of insulation is best at reducing the transfer of thermal energy from hot water.

Look at the diagram of Lily's experiment.



Look at her results.

cardboard		newspaper		bubble wrap	
time in seconds	temperature	time in seconds	temperature	time in seconds	temperature
0	83	0	85	0	85
60	67	60	81	60	81
120	75	120	79	120	79
180	71	180	76	180	77
240	68	240	72	240	75
300	65	300	68	300	73

- (a) The unit of temperature is missing from the tables.

Write down the unit of temperature.

..... [1]

- (b) Calculate the decrease in temperature in the 300 seconds for each beaker.

cardboard

newspaper

bubble wrap

[1]

(c) Before the investigation, Lily predicts,

'Bubble wrap is the best insulator.'

Is her prediction correct?

Explain your answer.

..... [1]

(d) In one of Lily's results tables, there is an anomalous result.

Circle the anomalous result in the table.

Give a reason for your answer.

..... [2]

(e) Lily improves her investigation.

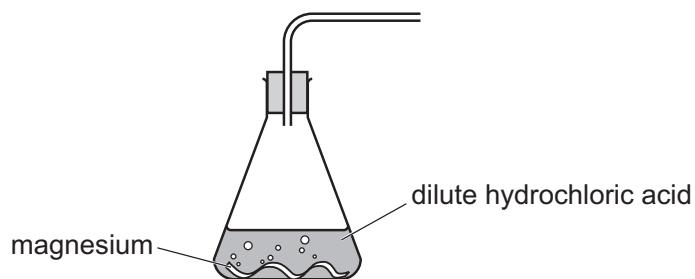
Suggest **two** improvements Lily makes to her investigation.

1

 2
 [2]

- 12 Mike investigates the reaction between magnesium and dilute hydrochloric acid.

Look at part of the equipment he uses.



- (a) Write down the name of the equipment Mike uses to **collect** the gas and **measure** the volume of the gas.

..... [1]

- (b) Mike measures the volume of gas made every 30 seconds until the reaction stops.

Describe how Mike makes his results more reliable.

.....
..... [1]

- (c) Mike writes a risk assessment for his investigation.

Write down **one** safety risk and describe how Mike reduces this risk.

safety risk

how Mike reduces this risk

.....
[2]

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The Periodic Table of Elements

Group									
1	2							7	8



Cambridge Lower Secondary Checkpoint

SCIENCE

0893/02

Paper 2

April 2023

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Markers were instructed to award marks. It does not indicate the details of the discussions that took place at a Markers' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the End of Series Report. Cambridge will not enter into discussions about these mark schemes.

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

Only the science is being assessed so answers do not need to be grammatically correct. Significant figures will be indicated in the question or in the mark scheme. Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>Underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working, provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

Question	Answer	Marks	Further Information
1(a)	DNA or deoxyribonucleic acid	1	Do not accept chromosome Ignore genes or genetic material or allele
1(b)(i)	2 white and 3 grey or 1 white and 4 grey or 0 white and 5 grey	1	

Question	Answer	Marks	Further Information
1(b)(ii)	any three from M1 (idea that) less predation of grey shrews or fewer grey shrews eaten or grey shrews are more difficult to hunt M2 grey shrews have better camouflage or (idea that) grey ones blend in with the ground better than white shrews M3 (idea that) more grey shrews survive or so grey shrews number increases M4 more grey shrews are able to breed or more grey shrews reproduce M5 pass grey (fur) gene onto next generation or pass advantageous gene onto next generation M6 (process) repeats for many generations	3	each correct answer = 1 mark Accept ora where appropriate Accept black or brown for grey throughout Accept idea that fewer grey shrews are killed Accept (idea that) easier to see the white shrews Accept pass on advantageous mutation to next generation

Question	Answer	Marks	Further Information
1(b)(iii)	any two from wear gloves (handling shrews) eye protection (face) mask or filter disinfect traps (between use) or sanitise (traps)	2	each correct answer = 1 mark Accept use nets (instead of hands) Accept glasses or goggles Ignore protective suit Accept wash hands after investigation or touching shrews

Question	Answer	Marks	Further Information
2(a)	2.8.7	1	Accept a diagram of the correct electronic structure
2(b)	Mg	1	Accept magnesium, but symbol takes precedence
2 (c)	any one from unreactive or inert gas at room temperature low melting point or low boiling point colourless	1	Do not accept little reaction Accept low density Note being a gas is not sufficient Accept has a higher boiling point than neon or has a higher melting point than neon Accept atom has 18 protons or proton number is 18 or atomic number is 18 or has a stable electronic structure or has a stable atomic structure or it has eight electrons in its outer shell Do not accept is coloured or colourful Do not accept is a metal or a halogen

Question	Answer	Marks	Further Information												
3(a)	<table><tr><th>part of plant</th><th>order of pathway</th></tr><tr><td>leaf</td><td>(5)</td></tr><tr><td>leaf xylem</td><td>4</td></tr><tr><td>root hair cell</td><td>1</td></tr><tr><td>root xylem</td><td>2</td></tr><tr><td>stem xylem</td><td>3</td></tr></table>	part of plant	order of pathway	leaf	(5)	leaf xylem	4	root hair cell	1	root xylem	2	stem xylem	3	1	all four correct for the mark
part of plant	order of pathway														
leaf	(5)														
leaf xylem	4														
root hair cell	1														
root xylem	2														
stem xylem	3														
3(b)	transpiration	1	Accept evaporation or evapotranspiration Ignore diffusion Do not accept sweating												
3(c)	reduces population size or population declines	1	Ignore they die or cannot grow as well or becomes extinct Ignore any reason given												

Question	Answer	Marks	Further Information
4(a)	W	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence
4(b)	(number of waves) 4	1	
4(c)	(time = 1 second) reinforce (time = 3 seconds) cancel each other or cancel out	2	each correct answer = 1 mark Accept constructive interference Accept (produces a) louder sound or amplify the sound or larger amplitude Accept destructive interference Accept (produces a) quieter sound or smaller amplitude

Question	Answer	Marks	Further Information
5(a)	respiration	1	Do not accept breathing
5(b)	photosynthesis	1	
5(c)	both release carbon dioxide or both release heat (into the surroundings) or both exothermic	1	Ignore they are both part of the carbon cycle Ignore they break down carbon (compounds) Do not accept both release carbon

Question	Answer	Marks	Further Information
5(d)	any two from ice caps melt or glaciers melt sea levels rise (more) flooding or (more) land under water or (more) cities under water drought or famine (more frequent) extreme weather, e.g. (more) typhoons/cyclones/tornadoes global warming extinction of plant (species) or extinction of animal (species) (increased number of) forest fires or wildfires	2	each correct answer = 1 mark Ignore reference to ozone and ozone depletion Note ice melts or snow melts unqualified is not sufficient Accept sea level changes Accept (more extreme) heat waves Ignore weather change or unpredictable weather Accept Earth or atmosphere becomes warmer Accept sea temperatures rise or ocean warming Accept temperature change or temperature increase Accept examples of extinction, e.g. danger of polar bears becoming extinct or mass extinction (of plants and/or animals) or coral bleaching or loss of habitats

Question	Answer	Marks	Further Information
6(a)	(as you go down the group atomic radius) increases or goes up	1	
6(b)	number in the range 20 to 45 (°C)	1	Accept a range of values providing the range is entirely within 20 to 45 °C Accept answer written in table, but answer line takes precedence
6(c)	giant (ionic)	1	Do not accept giant covalent Ignore lattice

Question	Answer	Marks	Further Information
7	buzzer variable resistor	2	each correct answer = 1 mark Accept rheostat Do not accept resistor on its own Do not accept variation resistor

Question	Answer	Marks	Further Information
8(a)	C B (E) D (F) A	3	all four letters in the correct order = 3 marks CB in correct order = 1 mark D in the middle = 1 mark A at the end = 1 mark
8(b)	rocks from the Earth and the Moon had the same properties or Moon rocks have a lower density than Earth rocks	1	Accept rocks from the Moon were similar to rocks on Earth or rocks from the Moon are same type as rocks on Earth or has the same components/composition as rocks on Earth or Moon and Earth contain the same minerals Do not accept Moon and Earth have an iron core
8(c)	any one from (idea that) the dense iron from the cores (of both planets) merged to create the Earth there are more dense rocks on the Earth the less dense rocks (were ejected and) formed the Moon	1	Accept ora Accept it was the less dense rocks that were ejected

Question	Answer	Marks	Further Information
9(a)	changes in seasons ☐ changes to the environment over time ☒ increased reproduction ☐ new diseases ☒ new food sources ☐	2	each correct tick = 1 mark three ticks and two correct = 1 mark three ticks and one correct = 0 marks four or five ticks = 0 marks Accept any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence
9(b)	(idea that) death rate greater than reproduction rate	1	Accept low reproduction rate and high death rate Accept death rate is increasing and the reproduction rate is decreasing Accept birth rate for reproduction rate Ignore reference to no reproduction

Question	Answer	Marks	Further Information
10(a)	displacement	1	more than one answer circled = 0 marks Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence
10(b)	copper + silver nitrate → copper nitrate + silver	1	Accept reactants in either order and products in either order Accept = for → Accept correct symbol equation (balanced or unbalanced), but names take precedence $\text{Cu} + 2\text{AgNO}_3 \rightarrow \text{Cu}(\text{NO}_3)_2 + 2\text{Ag}$

Question	Answer	Marks	Further Information
10(c)	copper + iron nitrate ☐ magnesium + zinc nitrate ☒ silver + magnesium nitrate ☐ iron + zinc nitrate ☐ magnesium is more reactive than zinc or magnesium is higher in the reactivity series than zinc or zinc is less reactive than magnesium or zinc is lower in the reactivity series than magnesium	2	each correct answer = 1 mark more than one answer ticked = 0 marks for the question incorrect answer ticked = 0 marks for the question Accept any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence Note magnesium is more reactive is not sufficient or it is more reactive than zinc is not sufficient Note the more reactive metal displaces the less reactive metal is not sufficient must refer to magnesium and zinc Do not accept magnesium is more reactive than zinc nitrate

Question	Answer	Marks	Further Information
11(e)	any two from use the same starting temperature use the same temperature of the surroundings or same room temperature use the same volume of water use a lid on each beaker use the same thickness of insulation repeat the investigation stir the water	2	each correct answer = 1 mark Accept use same mass of water or same amount of water or same level of water Accept cover the top of beaker or put a stopper on the beaker Ignore put a layer of oil on water so heat doesn't escape Accept same surface area of insulation Accept repeat the investigation and calculate the mean = 2 marks Accept greater variety of insulation Accept insulate the bottom of the beaker Accept smaller time intervals between readings Accept use a temperature sensor and a data logger Ignore a more accurate or precise thermometer

Question	Answer	Marks	Further Information
12(a)	(gas) syringe or measuring cylinder (inverted in a trough of water)	1	Accept graduated cylinder
12(b)	repeats (the investigation)	1	Ignore changing time intervals or using a different total time
12(c)	(safety risk) hydrochloric acid is corrosive (how the risk is reduced) wear goggles or gloves	2	each correct answer = 1 mark Note award the answers wherever they are seen Accept acid or hydrochloric or solution (in the flask) for hydrochloric acid Accept hydrochloric acid burns or hydrochloric acid is harmful or hydrochloric acid getting into eyes or hydrochloric acid is an irritant Note hydrochloric acid coming into contact with hands is not sufficient Ignore references to the gas being produced or explosions Do not accept hydrochloric acid is flammable Accept gloves or apron or lab coat or glasses Ignore facemask or protective gear